



DESIGN REPORT 1

BAJA SAEINDIA

CHENNAI - 2026

TEAM NAME : DIRT DEMON RACING

TEAM ID :BSC26105

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1.ABSTRACT

This report presents the Initial Design Review (DR-1) of the BAJA all-terrain vehicle developed by Team Dirt Demon Racing for SAE BAJA Chennai. The objective of the design is to develop a safe, reliable, and rule-compliant vehicle capable of performing efficiently under off-road conditions. Driver safety is prioritized through a rule-compliant roll cage, essential safety systems, and a robust chassis structure, while ensuring manufacturability within the team's fabrication capabilities. The suspension system is designed to provide adequate wheel travel, stability, and ride comfort on uneven terrain. The drivetrain utilizes the mandated stock engine to deliver sufficient torque for dynamic events such as acceleration, hill climb, and endurance. Hydraulic braking and optimized steering geometry are incorporated to ensure effective control and safety. All design decisions are based on justified assumptions derived from competition rules, expected terrain conditions, and practical constraints, forming a strong foundation for further development, fabrication, and testing.

INTRODUCTION

SAE BAJA is a premier engineering design competition organized by SAEISS that provides undergraduate students with hands-on experience in the complete product development cycle of an all-terrain vehicle. The competition requires teams to design, analyse, fabricate, and validate a single-seat off-road vehicle capable of performing reliably under harsh terrain and demanding dynamic conditions, while emphasizing safety, manufacturability, cost effectiveness, and teamwork.

Team Dirt Demon Racing is a college-based student team formed to apply theoretical engineering knowledge to real-world automotive challenges. The team works collaboratively across design, analysis, material selection, manufacturing, and testing domains. This Design Review (DR-1) report presents the initial conceptual design of the BAJA vehicle developed for SAE BAJA Chennai. The design approach considers rulebook constraints, expected off-road conditions, and fabrication capabilities, with safety, reliability, and simplicity as primary design priorities.

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1.1 Design assumption

- Vehicle layout was finalized using 1:1 scaled 2D drawings to accurately represent actual packaging and dimensions.
- Wheelbase (1751.56mm) was selected to achieve a balance between manoeuvrability in tight BAJA tracks and high-speed stability.
- Front and rear track widths (1461.94mm and 1461.95mm) were assumed to improve roll stability and reduce rollover tendency during cornering.
- Suspension geometry was positioned within the overall width limits to ensure no interference with chassis members during full jounce and rebound.
- Driver seating position was assumed such that the driver remains fully inside the roll cage envelope with adequate head, shoulder, and leg clearance as per BAJA safety intent.
- Steering wheel location and column angle were assumed for comfortable reach and control, considering average driver ergonomics.
- Pedal box position was fixed to allow full pedal travel without knee lock, ensuring safe braking and throttle operation.
- Powertrain (engine + CVT/drive system) was placed towards the rear to achieve rear-biased weight distribution, suitable for off-road traction.
- Adequate clearance was assumed between the engine, drivetrain, and chassis tubes to allow heat dissipation and ease of maintenance.
- Ground clearance was finalized considering off-road obstacles, suspension travel, and rulebook intent for BAJA terrain.
- Vehicle overall height (1567.91mm) was assumed to keep the center of gravity low while maintaining sufficient driver head clearance.
- Radiator and cooling components were positioned based on airflow availability and protection from debris.
- All subsystems were arranged assuming easy accessibility for service and inspection during technical scrutiny.
- Dimensional assumptions were cross-verified with the BAJA SAE India Rulebook to ensure compliance with packaging and safety constraints.
- The finalized 2D layout was frozen to act as the baseline reference for 3D CAD modelling, structural analysis, and subsystem design.

1.2 Design Justification

- A wheelbase of (1751.56 mm) was justified to provide sufficient longitudinal stability while still allowing the vehicle to negotiate tight turns on BAJA tracks.
- Track width of (1461.94mm and 1461.95 mm) was selected to improve lateral stability and reduce body roll during cornering and uneven terrain traversal.
- The front suspension layout was justified based on adequate clearance between wheels, shocks, and chassis members to avoid interference during maximum steering and suspension travel.
- Rear suspension and drivetrain placement were justified to support effective power delivery and maintain traction on loose surfaces.
- The driver seating position was justified to keep the driver fully enclosed within the roll cage, ensuring safety during rollover scenarios.
- Steering wheel and pedal locations were justified based on driver reachability and control, allowing natural hand and foot movement without fatigue.
- The low seating position was justified to reduce the overall center of gravity, improving handling and rollover resistance.
- Engine placement towards the rear was justified to achieve a rear-biased weight distribution, which is beneficial for acceleration and off-road performance.
- Adequate spacing between engine, exhaust, and chassis members was justified to reduce heat transfer and allow safe operation.
- Ground clearance was justified to ensure the vehicle can clear common BAJA obstacles such as rocks, ruts, and bumps without underbody contact.
- The overall vehicle height (1567.91mm) was justified to balance driver visibility, safety clearance, and a low center of gravity.
- Cooling system positioning was justified based on exposure to airflow while remaining protected from debris and impact.
- Subsystem placement was justified to allow ease of assembly, inspection, and maintenance, which is critical during endurance and technical inspection.
- All critical dimensions were justified by cross-checking with the BAJA SAE India rulebook, ensuring compliance with safety and packaging intent.
- The finalized 2D layout was justified as a reliable foundation for 3D CAD development, structural analysis, and subsystem optimization.

1.3 .Rulebook Reference & Compliance

Sl. No	Design Parameter	Rulebook Requirement	Proposed Design Value	Compliance Status	Remarks / Justification
1	Wheelbase	No specific max/min (should ensure stability & manoeuvrability)	1751.56 mm	Compliant	Optimized for off-road stability and turning radius
2	Front Track Width	≥ 1200 mm	1461.94 mm	Compliant	Exceeds minimum requirement for rollover stability
3	Rear Track Width	≥ 1200 mm	1461.95 mm	Compliant	Maintains uniform lateral stability
4	Ground Clearance	≥ 250 mm (at static condition)	291.84 mm	Compliant	Suitable for rough terrain and obstacles
5	Overall Width	≤ 1620 mm	1594.51 mm	Compliant	Ensures track manoeuvrability and rule compliance
6	Driver Seating	Driver must fit within roll cage envelope	Semi-reclined seating	Compliant	Improves ergonomics and CG while meeting safety rules
7	Driver Clearance	Helmet clearance from roll cage required	≥ 50 mm	Compliant	Verified in 2D side view layout
8	Steering System	Mechanical steering only	Rack & Pinion	Compliant	Simple, reliable, and rule-compliant
9	Suspension Type	Any, provided safety rules met	Independent suspension	Compliant	Provides better wheel articulation
10	Visibility	Driver must have clear forward visibility	Maintained	Compliant	Verified in side and front views

1.4 Architecture Selection & Rationale Table

System	Design Choice	Engineering Rationale	Rulebook Compliance
Vehicle Layout	Single-seat off-road BAJA prototype	Optimized for manoeuvrability and obstacle clearance	Compliant
Drivetrain	Rear Wheel Drive	Simplifies drivetrain and improves rear traction	Compliant
Powertrain Placement	Rear mounted	Better weight distribution and packaging	Compliant
Transmission	Chain-sprocket reduction	Cost-effective and easy to maintain	Compliant
Suspension	Independent suspension	Improves wheel articulation on uneven terrain	Compliant
Steering	Rack and pinion	Provides precise and mechanical steering control	Compliant
Braking	Hydraulic disc brakes	Ensures effective braking performance	Compliant
Chassis	Tubular space frame (AISI 1018)	High strength-to-weight ratio	Compliant
Seating	Semi-reclined single seat	Improves ergonomics and lowers CG	Compliant

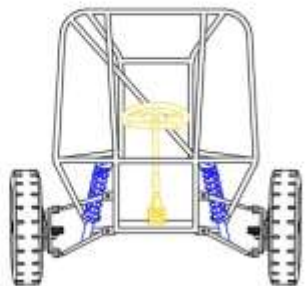
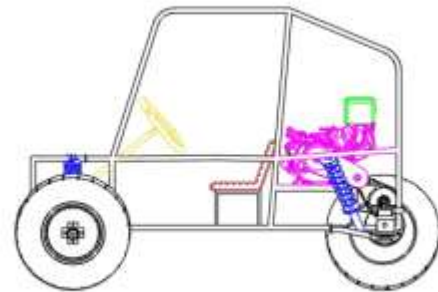
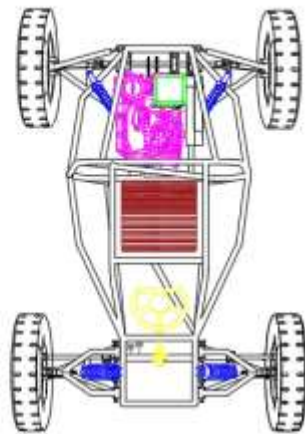
The overall vehicle architecture has been finalized by considering the operational requirements of the SAE BAJA off-road event, rulebook constraints, and manufacturability. A single-seat off-road prototype configuration has been selected to ensure better manoeuvrability and control on uneven terrain. The rear-wheel-drive layout with rear-mounted powertrain improves traction and optimizes weight distribution. Independent suspension architecture is adopted to enhance wheel articulation and ride comfort over obstacles. A mechanical rack-and-pinion steering system and hydraulic disc brakes are selected to ensure precise control and effective braking performance. The tubular space frame chassis using AISI 1018 material provides adequate strength while maintaining a favourable strength-to-weight ratio. The selected architecture is validated through scaled 2D layout drawings and found to comply with all applicable SAE BAJA rulebook requirements.

1.5.Driver Ergonomics Design Table

Sl. No	Parameter	Design Consideration	Proposed Value / Condition	Rulebook Compliance
1	Driver Type	Single driver	95th percentile male	Compliant
2	Seating Position	Semi-reclined posture	Reduces fatigue and lowers CG	Compliant
3	Seat Height from Ground	Driver eye level visibility	~600 mm	Compliant
4	Pedal Reach	Comfortable full pedal stroke	Adjustable pedal position	Compliant
5	Steering Wheel Reach	Easy reach without stretches	300–350 mm from chest	Compliant
6	Helmet Clearance	Vertical clearance to roll cage	≥ 50 mm	Compliant
7	Shoulder Clearance	Lateral clearance	≥ 75 mm	Compliant
8	Knee Clearance	Forward knee movement	No interference with steering	Compliant
9	Field of View	Forward and side visibility	Unobstructed	Compliant
10	Safety Envelope	Driver inside roll cage	Fully enclosed	Compliant

The driver ergonomics and seating position have been designed to ensure safety, comfort, and effective vehicle control during endurance and dynamic events. A semi-reclined seating posture is selected to reduce driver fatigue and lower the vehicle's center of gravity. The layout accommodates a 95th percentile male driver to ensure inclusivity while maintaining adequate safety clearances as per the SAE BAJA rulebook. Proper reachability to the steering wheel and pedals is ensured without excessive stretching, and sufficient helmet, shoulder, and knee clearances are maintained within the roll cage envelope.

1.6 Subsystem Placement & Packaging



- STEERING
- ENGINE
- SUSPENSION
- SEAT
- FUEL TANK
- PEDAL

Subsystem Placement table

Sl. No	Subsystem	Placement Location	Packaging Consideration	Compliance / Remarks
1	Engine	Rear section of vehicle	Improves traction and weight distribution	Compliant with rulebook
2	Transmission (Chain–Sprocket)	Rear drivetrain area	Short chain length and easy maintenance	Packaging verified
3	Final Drive	Rear axle line	Aligns with wheel centreline	No interference observed
4	Suspension System	Front and rear corners	Adequate suspension travel without fouling	Verified in side view
5	Steering System	Front section	Clear steering link movement ensured	Mechanical and compliant
6	Brake System	Near wheel hubs	Short hydraulic lines for efficiency	Safe routing ensured
7	Fuel Tank	Protected central-rear zone	Lowers CG and ensures safety	Within roll cage envelope
8	Driver Seat	Central position	Balanced weight distribution	Ergonomics verified
9	Roll Cage	Surrounding driver	Maintains safety envelope	Rulebook compliant
10	Electrical / Controls	Accessible locations	Easy serviceability	Properly packaged

The placement and packaging of all major subsystems are finalized to ensure proper weight distribution, safety, and ease of maintenance while complying with SAE BAJA rulebook requirements. The powertrain components are positioned in the rear section to improve traction and optimize longitudinal weight balance. Suspension, steering, and braking systems are arranged to allow full functional movement without interference throughout their operating range. Critical components such as the battery or fuel tank are placed within the protected zone of the roll cage to enhance safety. All subsystem placements are validated using scaled 2D layout drawings, ensuring adequate clearances, accessibility, and serviceability during fabrication and operation.

1.7 Accessibility & Serviceability

Accessibility and Serviceability (DR1 Submission)

- Major subsystems such as engine, transmission, suspension, and braking components are positioned to allow easy visual inspection without dismantling the chassis.
- The engine compartment layout provides sufficient space around the engine for routine maintenance activities such as oil change, chain inspection, and belt/sprocket adjustment.
- Suspension mounting points are kept accessible to allow quick damper removal, preload adjustment, and replacement during event time constraints.
- Brake components such as master cylinder, brake lines, and callipers are placed to allow easy bleeding, inspection, and repair.
- The steering system layout ensures accessibility to tie rods, steering rack/column joints for alignment checks and tightening.
- Electrical components (battery, wiring harness, kill switch connections) are placed in reachable locations to enable quick troubleshooting and safe isolation during emergencies.
- Cooling system components are positioned to allow easy cleaning of radiator/fins and hose inspection without disturbing other subsystems.
- Fasteners used in critical areas are oriented to allow tool access using standard hand tools, minimizing service time.
- Adequate clearance is maintained between subsystems to avoid obstruction during assembly and disassembly operations.
- The layout allows quick removal of wheels for tire replacement and suspension access during dynamic events.
- Subsystems are grouped logically to reduce maintenance complexity and improve service efficiency.
- The overall vehicle layout was reviewed to ensure that routine servicing can be performed within competition time limits.
- Accessibility and serviceability considerations were validated during the 2D layout stage to avoid major redesign during fabrication.

1.8 Compliance Verification Summary

S. No	Rule Category	Rule Requirement (As per Rulebook)	Design Specification (Proposed Vehicle)	Compliance Status	Remarks / Justification
1	Overall Dimensions	Vehicle dimensions shall be within rulebook limits	Overall length $\approx 2380.18\text{mm}$, width = 1594.51 mm , height $\approx 1567.91\text{ mm}$	Compliant	Verified using scaled 2D layout
2	Track Width	Track width within permissible range	Front $\approx 1461.94\text{mm}$ Rear $\approx 1461.95\text{mm}$	Compliant	Measured at wheel centrelines
3	Wheelbase	Wheelbase as per rule constraints	$\approx 1751.56\text{ mm}$	Compliant	Ensures stability and manoeuvrability
4	Ground Clearance	Minimum ground clearance to be maintained	$\approx 291.84\text{ mm}$ (static)	Compliant	Suitable for off-road terrain
5	Driver Ergonomics	Driver must fit safely with clearances	Seat, steering and pedal layout validated	Compliant	Based on 95th percentile driver
6	Roll Cage Envelope	Driver must fit inside roll cage envelope	All clearances maintained in 2D layout	Compliant	Head, shoulder, leg clearances checked
7	Steering System	Mechanical steering with safe geometry	Rack and pinion system adopted	Compliant	Proper Ackermann geometry ensured
8	Suspension	Independent suspension allowed	Independent front & rear suspension	Compliant	Adequate wheel travel provided
9	Brake System	Dual hydraulic braking system required	Hydraulic disc brakes on all wheels	Compliant	Meets safety requirements

10	Powertrain Layout	Safe placement of engine & drivetrain	Rear-mounted engine with chain drive	Compliant	Guarded and isolated from driver
11	Accessibility	Systems must be serviceable	Engine, brakes, suspension accessible	Compliant	Verified during layout review
12	Safety Controls	Kill switches must be accessible	Driver-reachable kill switch provided	Compliant	Location considered in layout



Outcome of DR-1 Submission

- Finalized and frozen 1:1 scaled 2D vehicle layout
- Verified driver ergonomics and safety clearances
- Confirmed rulebook compliance for all major dimensions
- Established baseline architecture for 3D CAD modelling
- Validated subsystem packaging and accessibility
- DR-1 approval enables transition to detailed design & analysis (DR-2)





THANKYOU